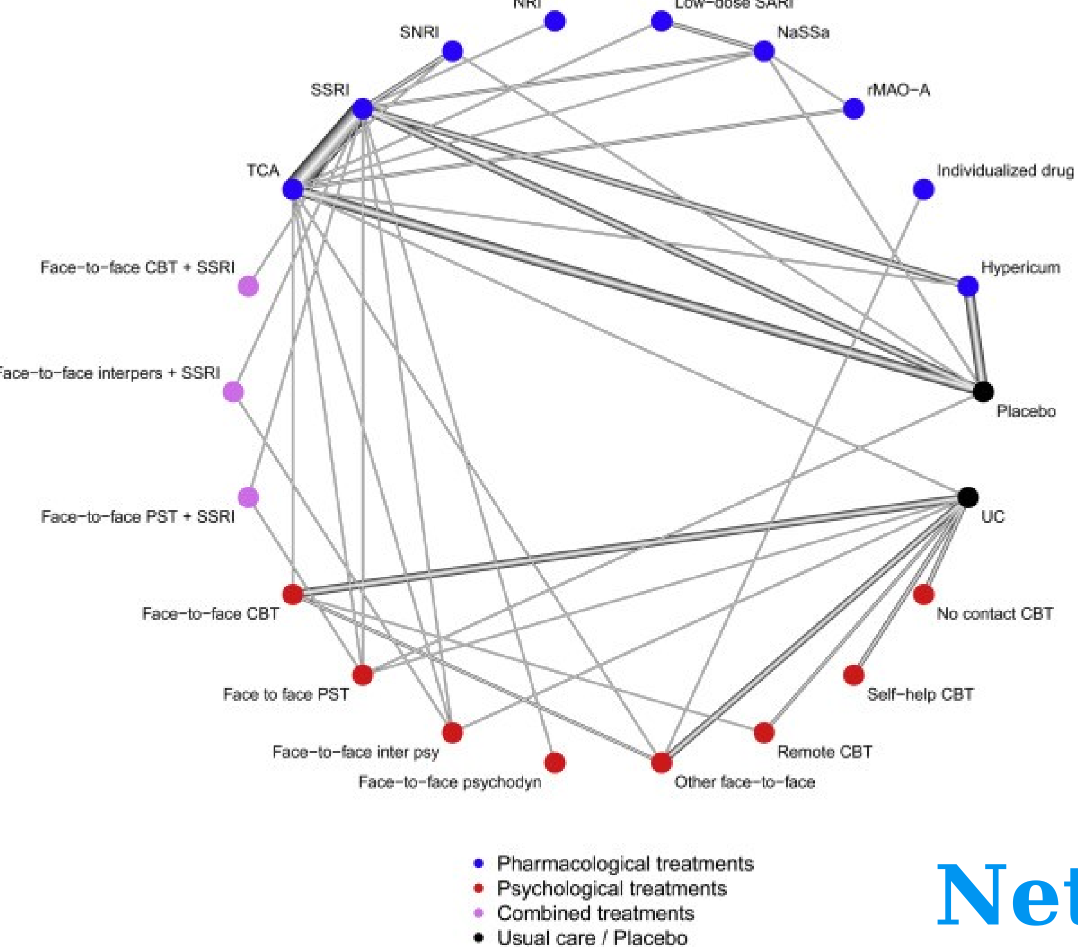
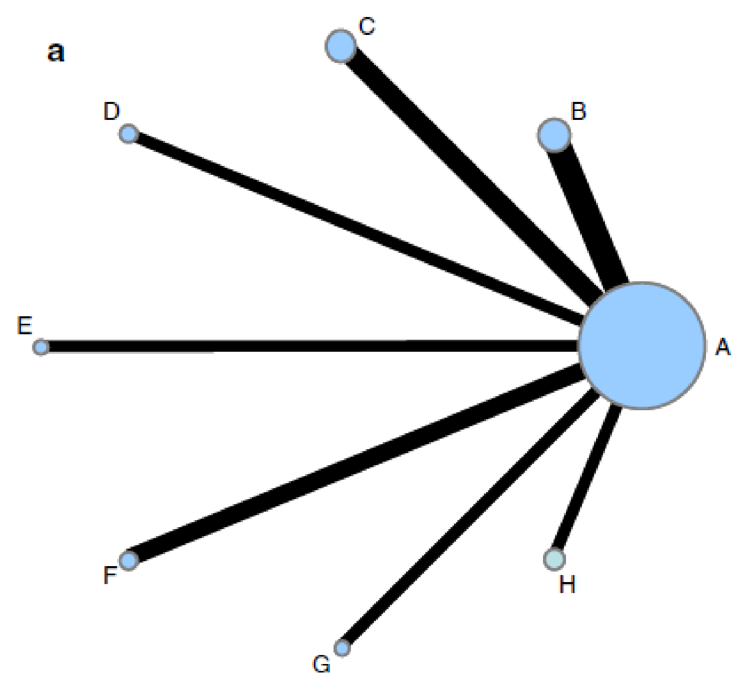
Application and extensions: Network meta-analysis (NMA)

Prof. Wim Van den Noortgate
Faculty of Psychology and Educational Sciences
KU Leuven – University of Leuven, Belgium
Wim.VandenNoortgate@kuleuven.be





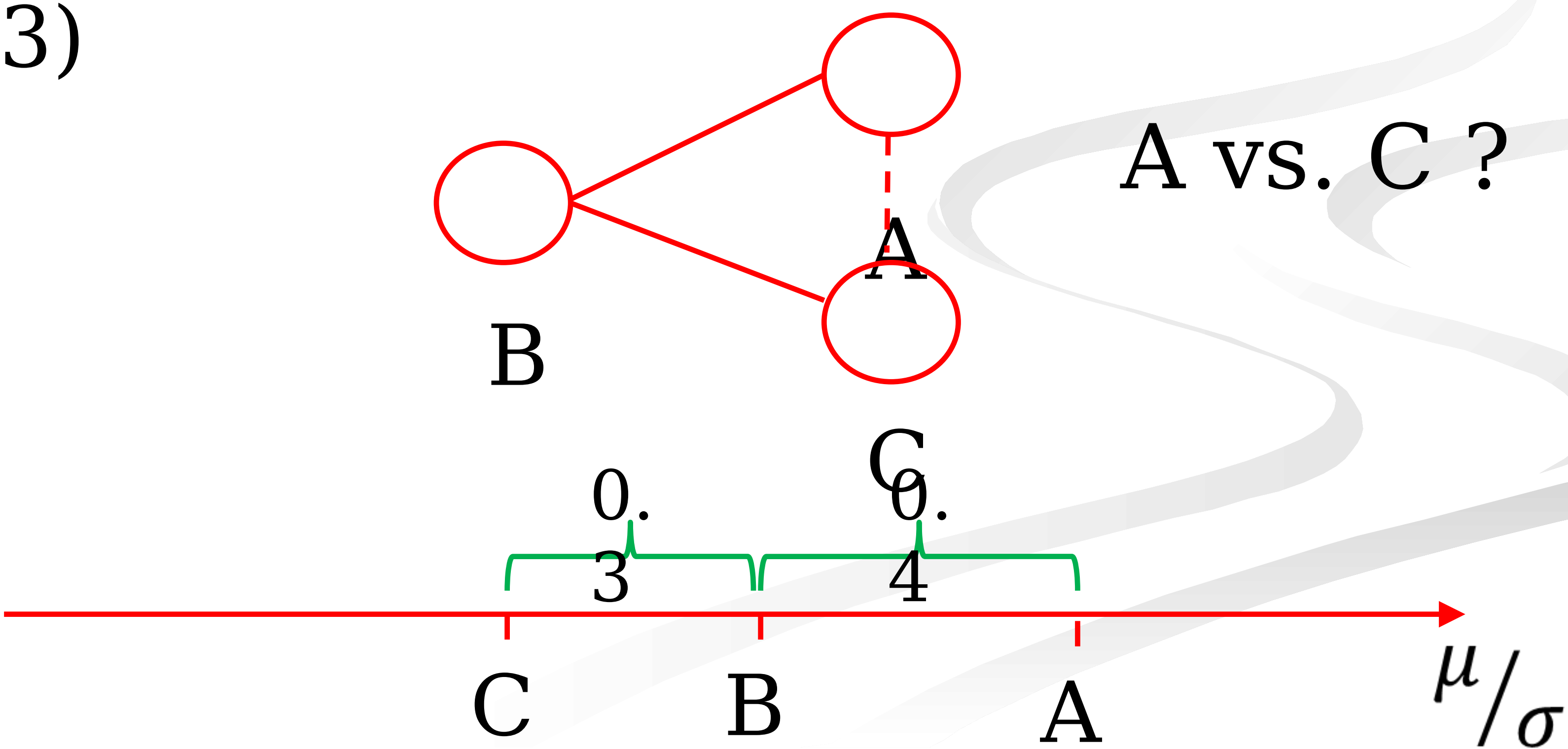
Network graphs

Linde, K., Rücker, G., Schneider, A., & Kriston, L. (20
Questionable assumptions hampered interpretation o
meta-analysis of primary care depression treatments
Clinical Epidemiology, 71, 86-96.

Network meta-analysis
(=multiple treatment meta-analysis)

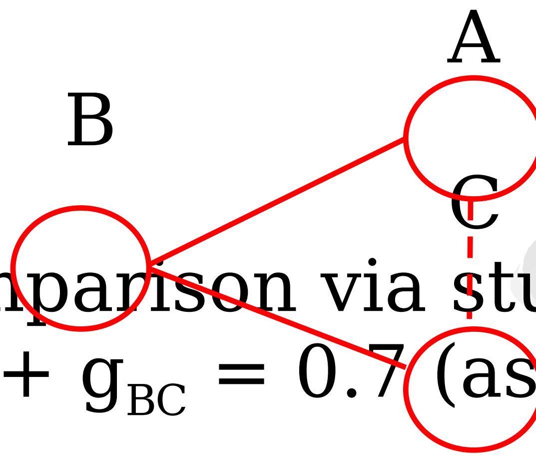
Study 1: Treatment A vs. B (e.g., g_{AB} = 0.4)

Study 2: Treatment B vs. C (e.g., g_{BC} = 0.3)



Network meta-analysis
(=multiple treatment meta-analysis)

Study 1: Treatment A vs. B (e.g., $g_{AB} = 0.4$)
Study 2: Treatment B vs. C (e.g., $g_{BC} = 0.3$)



A vs. C ?

⇒ Indirect comparison via study 1 and 2

$g_{AC} = g_{AB} + g_{BC} = 0.7$ (assuming **transitivity**)

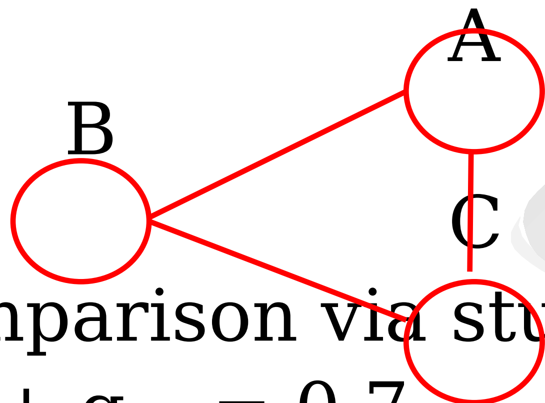
$var(g_{AC}) = var(g_{AB}) + var(g_{BC})$

Direct vs. indirect evidence

Study 1: Treatment A vs. B (e.g., $g_{AB} = 0.4$)

Study 2: Treatment B vs. C (e.g., $g_{BC} = 0.3$)

Study 3: Treatment A vs. C (e.g., $g_{AC} = 0.9$)



A vs. C ?

⇒ Indirect comparison via study 1 and 2

$$g_{AC} = g_{AB} + g_{BC} = 0.7$$

$$var(g_{AC}) = var(g_{AB}) + var(g_{BC})$$

Direct vs. indirect evidence

Study 1: Treatment A vs. B (e.g., $g_{AB} = 0.4$)

Study 2: Treatment B vs. C (e.g., $g_{BC} = 0.3$)

Study 3: Treatment A vs. C (e.g., $g_{AC} = 0.9$)

⇒ <u>Direct</u> comparison of A vs. C via study 3

$$g_{AC} = 0.9$$

⇒ <u>Indirect</u> comparison via study 1 and 2

$$g_{AC} = g_{AB} + g_{BC} = 0.7$$

$$\mathrm{var}(g_{AC}) = \mathrm{var}(g_{AB}) + \mathrm{var}(g_{BC})$$

Why using indirect evidence?

- increased precision by combining direct and indirect evidence (by using more information)
- compare treatments that have not been compared in a direct way (most studies compare treatments with a placebo or active control)

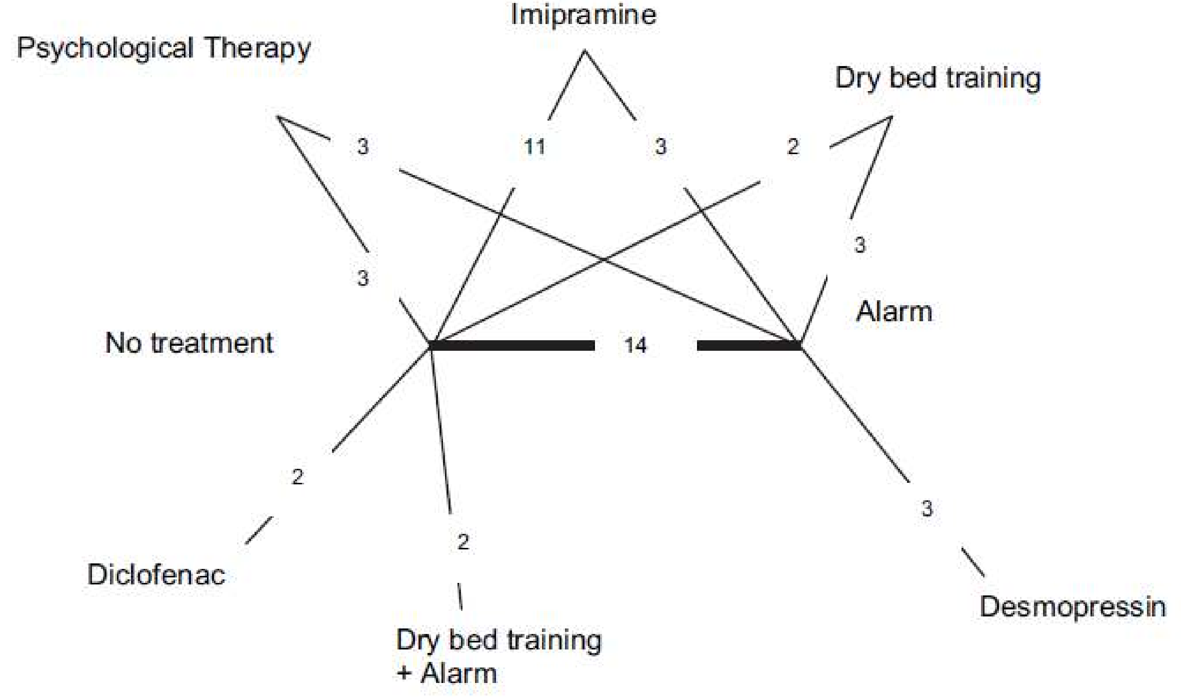
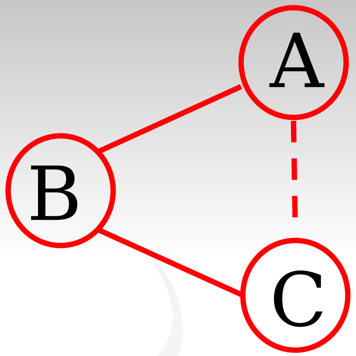


Fig. 1. Network of evidence for childhood nocturnal enuresis treatments. Each black line joining two treatments represents a direct head-to-head comparison.

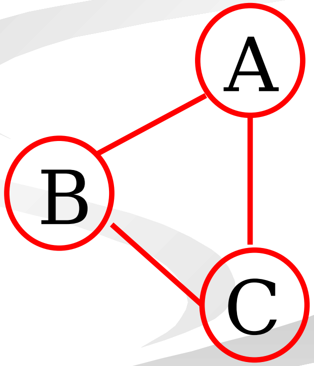
D.M. Caldwell et al. / Journal of Clinical Epidemiology 63 (2010) 875–882

Network MA

1. For open networks: use results of MA on AB-studies and MA on BC-studies to obtain an **indirect estimate** of A vs. C



2. For closed networks: calculate a weighted average of the indirect estimate from above and the direct estimate from MA of AC-studies as a **mixed estimate**



Note: in both cases **transitivity** assumed!

But … transitivity assumption $(\delta_{AC} = \delta_{AB} + \delta_{BC})$

- Study 1: BMW much faster than Range Rover

- Stu ve than Porsche

- Co

n I

Transitivity assumption

$\delta_{AC} = \delta_{AB} + \delta_{BC}$

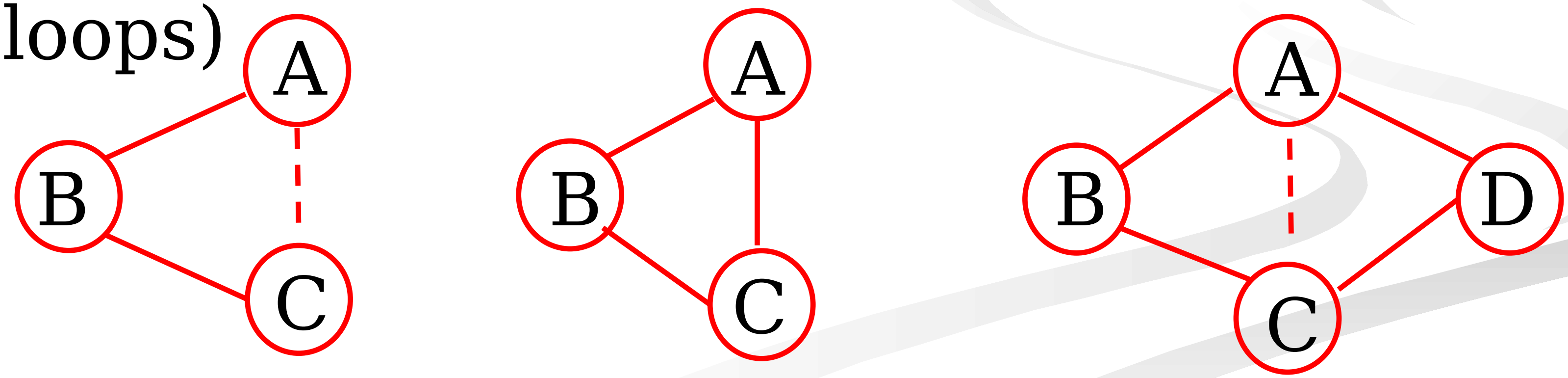
- treatment B is doing equally good in the AB-comparison studies and in the BC-comparison studies
- 'effect modifiers' (participant, study, treatment characteristics that influence performance of treatment) are distributed similarly in AB-comparison studies and BC-comparison studies

⇒ problem if Rover-BMW comparisons more often done in muddy conditions and/or Porsche-Rover comparisons more often done with old Rover

⇒ problem if new drug study includes typically patients for which traditional treatments do not work anymore

Consistency

- Transitivity assumption can not be tested statistically, to be judged by experts
- Violation of transitivity assumption can lead to inconsistencies in data (for closed loops only, so not for first network below!)
- Inconsistencies can be tested (for closed loops)



Inconsistent findings?

Study 1: Treatment A vs. B (e.g., $g_{AB} = 0.4$)
Study 2: Treatment B vs. C (e.g., $g_{BC} = 0.3$)
Study 3: Treatment A vs. C (e.g., $g_{AD} = 0.9$)

$g_{AC} = 0.7$

Possible explanations?

1. Unreliability (random variation within studies)
2. Heterogeneity between studies
3. Real inconsistency

Inconsistent findings?

Study 1: Treatment A vs. B (e.g., $g_{AB} = 0.4$)

Study 2: Treatment B vs. C (e.g., $g_{BC} = 0.3$)

$\Rightarrow g_{AC} = \boxed{0.7}$

Study 3: Treatment A vs. D (e.g., $g_{AD} = -0.2$)

Study 4: Treatment D vs. C (e.g., $g_{DC} = -0.4$)

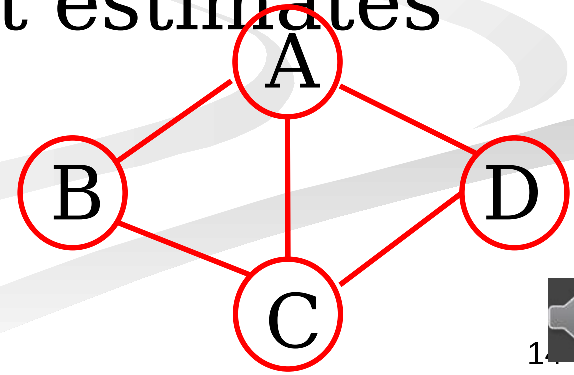
$\Rightarrow g_{AC} = \boxed{-0.6}$

Possible explanations?

1. Unreliability (random variation within studies)
2. Heterogeneity between studies
3. Real inconsistency

Checking transitivity/consistency

1. Check distribution of effect modifiers over types of studies
2. Test ‘inconsistency factor’ = difference between direct and indirect estimate (Bucher method)
3. Gleser & Olkin approach to test consistency of multiple estimates (e.g., one direct and two indirect estimates
4. Estimate and test model inconsistency



A linear mixed model approach

- $$MD_{ijk} = (\mu_i - \mu_j) + \eta_{ik} + \eta_{jk} + \xi_{ij} + r_{ijk}$$

$$r_{ijk} \sim N\left(0, \sigma^2_{r_{ijk}}\right)$$ **Sampling variance**

$$\eta_{ik}, \eta_{jk} \sim N\left(0, \sigma^2_{\tau}\right.$$ **Heterogeneity**

$$\xi_{ij} \sim N\left(0, \sigma^2_{\xi}\right)$$ **Inconsistency**

What is added to the expected mean difference if treatment *i* and treatment *j* are compared in the same study

Lumley, T. (2002). Network meta-analysis for indirect treatment comparisons. *Statistics in Medicine, 21*, 2313-2324.

A linear mixed model approach

- $$MD_{ijk} = \left(\mu_2 X_{2ijk} + \mu_3 X_{3ijk} + \mu_4 X_{4ijk} + \mu_5 X_{5ijk}\right) + \eta_{ik} + \eta_{jk} + \xi_{ij} + r_{ijk}$$

i vs. j	X_2	X_3	X_4	X_5	MD
2 vs. 3	1	-1	0	0	0.66
2 vs. 5	1	0	0	-1	0.43
1 vs. 3	0	-1	0	0	-0.23

But …

Tests of inconsistency may be underpowered due to:

1. Small number of studies giving (direct) evidence
2. Large heterogeneity

⇒ Lack of significance is no proof that there is consistency!

Dealing with inconsistency

1. Use of direct effect estimates may be safer than indirect or mixed estimates
2. Try to explain inconsistency (look at effect modifiers)
3. Account for additional uncertainty due to inconsistencies (cfr. σ_{ξ}^{2} from linear mixed model)

Remarks

1. Width of the network?
    1. Broader network = more evidence, but also more complex & much more comparisons
    2. Do sensitivity analyses!
2. If one study includes multiple direct effect estimates (= multi-arm studies), we should take into account that these are correlated (overlap in information)